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Software Engineering

SE101 | Suggested sequence · Semester 3 | Student-authored study guide

How to use these notes

Requirements, decomposition, version control, testing, review, deployment, and maintainable systems. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · Requirements and design

Record users, goals, constraints, and acceptance criteria. Decompose work into cohesive modules with explicit interfaces. Document trade-offs and validate assumptions with small prototypes.

Worked example

```
Acceptance criterion: “A signed-in owner can download their PDF, while a different account receives not-found.”
```

Check your understanding

- What makes an acceptance criterion testable?
- What is cohesion?

MODULE 02 · Implementation and verification

Use version control, small changes, reviews, automated tests, and clear error handling. Test behavior at unit, integration, and end-to-end levels according to risk.

Worked example

```
A regression test should fail before a bug fix and pass after it.
```

Check your understanding

- What does a regression test protect?
- Why review small changes?

MODULE 03 · Delivery, security, and maintenance

Automate repeatable builds and deployments. Apply least privilege, validate inputs, protect secrets, monitor failures, and plan backups and recovery. Treat maintenance and accessibility as ongoing engineering work.

Worked example

```
Store credentials in the deployment secret manager, not source code or client bundle.
```

Check your understanding

- What is least privilege?
- Why test a restore rather than only making backups?

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.